

# Kratom exposures among older adults reported to U.S. poison centers, 2014–2019

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## Abstract

**Background:** In recent years, use of the herbal supplement kratom has increased in the United States. The reasons for use include pain relief, particularly as a substitute for opioids.

**Objectives:** To describe epidemiologic trends in kratom-related exposures among older adults reported to U.S. poison centers.

**Design:** Retrospective analysis of American Association of Poison Control Center's National Poison Data System (NPDS).

**Setting:** Data from all U.S. poison centers from 2014 to 2019 were examined.

**Participants:** Kratom exposure cases involving adults aged 18 and older. Kratom cases were identified by product and NPDS generic codes. Non-human and information-only calls were excluded. Data were examined for all calls for exposures among adults, with a focus on older adults aged 60–69 years and above 70 years.

**Measurements:** Descriptive analyses were used to characterize individual demographic, exposure information, clinical effects, and medical outcomes associated with kratom exposures among older adults. Comparisons across age groups (18–59, 60–69, and 70+ years) were made using Fisher's exact tests.

**Results:** Among 3484 kratom-related exposures reported between 2014 and 2019, 4.6% ( $n = 162$ ) were among adults over 60 years. The number of kratom-related exposures increased over time. Most cases originated with calls from healthcare facilities (81.1%) and involved kratom as a single ingested (63.0%). The reason for most ingestions was intentional (74.5%). One in five exposures among adults aged 70 and older involved an adverse reaction (e.g., drug

interaction; 21.9%), compared with 12.3% among ages 60–69 and 9.6% among ages 18–59 years. Neurological and cardiovascular clinical effects were observed. Twenty-three deaths were observed among older adults.

**Conclusion:** Healthcare providers and older adult patients should be aware of the potential risks of kratom use, including medication interactions and falls. When reviewing medication lists, providers should query this population for all medications and substances being used, especially in people being treated for pain.

#### KEYWORDS

exposures, kratom, poison center, substance

## INTRODUCTION

Kratom (*Mitragyna speciosa*), an herbal supplement native to Southeast Asia, has been used for centuries in socioreligious ceremonies, as an aid for laborers to combat fatigue, as an opium substitute, and for other medical purposes.<sup>1</sup> Kratom has both stimulant and opioid-like properties, and the effects of consuming kratom are dose-dependent: smaller doses produce stimulatory effects, whereas larger doses produce effects that resemble opiates.<sup>1–3</sup> Kratom is available in different forms, including raw leaves, capsules, powder, and extracts,<sup>1</sup> which are sold in U.S. retail establishments and through online storefronts; however, the manufacturing, sale, or possession of kratom products are not currently regulated by the Food and Drug Administration (FDA) or Drug Enforcement Administration in the United States. The FDA has issued several statements, advisories, or alerts associated with kratom, including an August 2016 proposal to temporarily list active ingredients in kratom in Schedule I, which was withdrawn in October 2016 after receiving public comments.<sup>4</sup> Because it is not classified as a dietary supplement in the United States, kratom (and ingredients) is not subject to regulatory standards and requirements set forth by the Dietary Supplement Health and Education Act of 1994, which establishes marketing, labeling, and safety requirements for dietary supplements and dietary ingredients in the United States.

In the past, kratom has not been widely used in the United States. In 2019, less than 1% of U.S. adults reported the use of kratom.<sup>5,6</sup> However, anecdotally, interest seems to be growing. Although population-based data indicate the highest prevalence of past-year kratom use among young adults, with adults aged 55–64 and over 65 comprising of only 4.6% and 2.0% of kratom users,<sup>5</sup> online surveys suggest that up to 17%–27% of kratom users are over the age 50 years,<sup>7,8</sup> although this could be an over- or underestimate due to the survey mode.

### Key points

- Kratom (*Mitragyna speciosa*) is an herbal supplement and an emerging substance of concern with potential for abuse.
- Our findings indicate that from 2014 to 2019, U.S. poison centers have observed a rapid increase in kratom exposures, with nearly 5% of kratom-related calls among adults over age 60 years.
- Clinical outcomes associated with kratom exposure among older adults include severe effects (49.3% of cases were life-threatening or resulted in significant disability or disfigurement) and eight deaths.

### Why Does this Paper Matter?

This is the first study to examine exposures to an emerging drug, kratom (*M. speciosa*), which have increased among older adults from 2014 to 2019. Clinical outcomes associated with kratom exposure include severe effects, including death. Older adults and healthcare providers should be aware of this substance and its associated risks.

Reasons for kratom use vary by age.<sup>7,9</sup> Kratom has been as identified as an opioid substitute,<sup>7,10–15</sup> and such use is supported by patients and advocates.<sup>8,16–18</sup> Several recent surveys identified self-treatment of acute or chronic pain as the primary reason for kratom use.<sup>7,8,19</sup> Use for pain management may be of particular interest for older adults seeking pain control alternatives to opioids. Findings from one online survey of over 8000 adults recruited through social media in 2016 suggested that

older adults (aged 60 years and above) were significantly more likely to report taking kratom for prescription medicine dependency or for a medical condition leading to acute or chronic pain than people aged 18–20 years.<sup>7</sup> However, data from another online survey in 2017 did not find associations between age and use of kratom for opioid reduction purposes.<sup>19</sup>

Kratom use has been associated with deleterious mental and physical health effects. Negative effects of acute kratom exposure include chronic insomnia, cognitive impairment, psychosis, tremors, seizures, and liver toxicity.<sup>20–24</sup> Withdrawal from kratom also results in symptoms, including tremors, muscle spasms and jerky limb movements, restlessness, insomnia, aggressive behavior and mood swings, diarrhea, runny nose, teary eyes, body aches and pains, and mood swings,<sup>25,26</sup> which have been reported to be present among approximately 10% of users.<sup>7,27,28</sup>

Kratom may present particular risks for older adults. The high prevalence of prescription drug use among older adults, together with pre-existing conditions opens the door for potential negative medication interactions. Further, kratom side effects, such as dizziness and vertigo,<sup>7,8</sup> could contribute to increased fall risk in an already high risk older adult population, which would be consistent with the increased risk of falls associated with opioid and other psychoactive medications.<sup>29,30</sup> Additionally, to our knowledge, no study has examined the pharmacokinetics or pharmacodynamics of kratom in the older adult population.

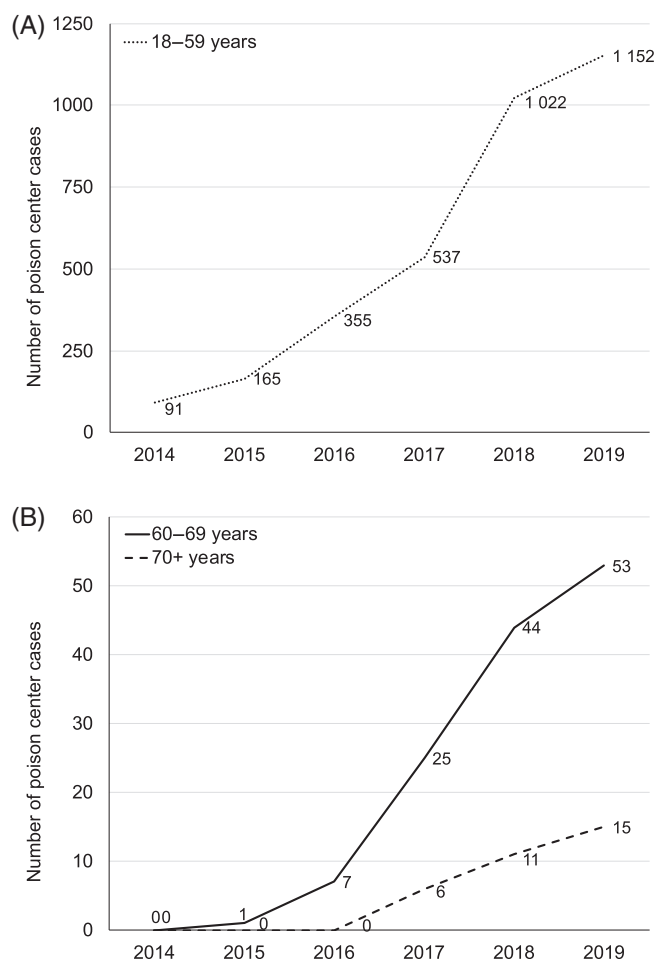
As interest in and access to kratom seem to be increasing in the United States, there has been a corresponding increase in calls to poison control centers, some associated with serious medical outcomes, including deaths.<sup>20,31</sup> To date, no study has examined kratom exposures in the older adult population. Studies that focus on young and middle-aged adults may not represent the complex health history and medications of older adults. The purpose of this research is to provide a description of kratom exposures among adults over age 60 reported to U.S. poison centers from 2014 to 2019.

## METHODS

Data related to all kratom exposures reported to U.S. poison centers between January 1, 2014, and December 31, 2019, were obtained from the American Association of Poison Control Centers (AAPCC) National Poison Data System (NPDS).<sup>32</sup> The NDPS query included codes for *plants-mitragyna* (product code 4271683) and *kratom* (generic code 0310130). Non-human

(e.g., animal) and information-only calls were excluded; cases with coingestants (more than one type of substance reported in the exposure) were included.

Demographic characteristics of cases, chronicity of exposure (acute, chronic, unknown), exposure sites (residence, other, unknown), reason for exposure (intentional, unintentional, adverse reaction, other), exposure route (ingestion, inhalation, other), substance formulation (solid, powder, liquid, other, unknown), clinical effects, and clinical outcomes, were summarized for all cases. Clinical outcomes included no effect, minor effect (minimal symptoms that typically resolve rapidly), moderate effect (pronounced, more prolonged symptoms or those of a more systemic nature than minor symptoms), major effect (life-threatening or symptoms resulting in significant residual disability or disfigurement), and death.<sup>32</sup> Data were described for all calls for exposures among older adults aged 60–69 years and above 70 years. We used a difference-in-difference analysis to compare the number of cases per year across age groups (18–59,



**FIGURE 1** Annual number of kratom-related exposures among adults aged 18–59 years (A) and 60–69 and above 70 years (B) reported to U.S. poison centers, 2014–2019

**TABLE 1** Demographic and case characteristics of kratom exposures reported to U.S. poison centers, 2014–2019

|                               | 18–59 years<br>( <i>n</i> = 3322) | 60–69 years<br>( <i>n</i> = 130) | 70 years and<br>above ( <i>n</i> = 32) | All cases<br>( <i>N</i> = 3484) | <i>p</i> -value   |
|-------------------------------|-----------------------------------|----------------------------------|----------------------------------------|---------------------------------|-------------------|
| Caller site                   |                                   |                                  |                                        |                                 | 0.41              |
| Healthcare facility           | 81.3                              | 76.9                             | 78.1                                   | 81.1                            |                   |
| Other (e.g., residence)       | 18.7                              | 23.1                             | 21.9                                   | 18.9                            |                   |
| Type of exposure              |                                   |                                  |                                        |                                 | 0.01              |
| Single substance              | 62.5                              | 74.6                             | 71.9                                   | 63.0                            |                   |
| Multiple substances           | 37.5                              | 25.4                             | 28.1                                   | 37.0                            |                   |
| Chronicity                    |                                   |                                  |                                        |                                 | 0.56              |
| Acute                         | 70.5                              | 67.7                             | 75.0                                   | 70.4                            |                   |
| Chronic                       | 17.4                              | 22.3                             | 18.8                                   | 17.6                            |                   |
| Unknown                       | 12.1                              | 10.0                             | 6.3                                    | 11.9                            |                   |
| Gender                        |                                   |                                  |                                        |                                 | <0.001            |
| Female                        | 30.9                              | 49.2                             | 50.0                                   | 31.8                            |                   |
| Male                          | 69.0                              | 50.8                             | 50.0                                   | 68.2                            |                   |
| Unknown                       | 0.1                               | 0.0                              | 0.0                                    | 0.1                             |                   |
| Exposure site                 |                                   |                                  |                                        |                                 | 0.70              |
| Residence                     | 87.6                              | 94.6                             | 93.8                                   | 87.9                            |                   |
| Other                         | 6.2                               | 4.6                              | 3.1                                    | 6.1                             |                   |
| Unknown                       | 6.2                               | 0.8                              | 3.1                                    | 6.0                             |                   |
| Reason                        |                                   |                                  |                                        |                                 | 0.03 <sup>a</sup> |
| Intentional                   | 74.9                              | 64.6                             | 68.8                                   | 74.5                            |                   |
| Unintentional                 | 7.0                               | 12.3                             | 9.4                                    | 7.3                             |                   |
| Adverse reaction              | 9.6                               | 12.3                             | 21.9                                   | 9.8                             |                   |
| Other                         | 4.9                               | 6.2                              | 0.0                                    | 4.9                             |                   |
| Unknown                       | 3.5                               | 4.6                              | 0.0                                    | 3.5                             |                   |
| Exposure route                |                                   |                                  |                                        |                                 | 0.58              |
| Ingestion                     | 56.1                              | 70.0                             | 78.1                                   | 56.9                            |                   |
| Inhalation                    | 2.0                               | 0.8                              | 3.1                                    | 2.0                             |                   |
| Ingestion w/ inhalation       | 2.2                               | 0.8                              | 0.0                                    | 2.1                             |                   |
| Other (e.g., rectal, vaginal) | 0.1                               | 0.0                              | 0.0                                    | 0.1                             |                   |
| Unknown                       | 39.6                              | 28.5                             | 18.8                                   | 39.0                            |                   |
| Substance formulation         |                                   |                                  |                                        |                                 | 0.14 <sup>b</sup> |
| Solid (tablet/capsule)        | 39.8                              | 37.7                             | 56.3                                   | 39.8                            |                   |
| Liquid                        | 7.6                               | 5.4                              | 6.3                                    | 7.5                             |                   |
| Powder/granules               | 15.0                              | 20.0                             | 18.8                                   | 15.2                            |                   |
| Other (aerosol, cream, patch) | 17.2                              | 16.2                             | 3.1                                    | 17.0                            |                   |
| Unknown                       | 20.4                              | 20.8                             | 15.6                                   | 20.4                            |                   |

Note: Values represent percentages. *p*-value indicates results from Fisher's exact tests across age groups.

<sup>a</sup>*p*-value reflects comparison between reasons for use; cases with reason "unknown" not included.

<sup>b</sup>*p*-value reflects comparison between solid substance formulation versus non-solid formation (all other categories combined).

60–69, and 70+ years). Comparisons in categorical variables (e.g., demographics) across age groups were made using Fisher's exact tests. The level of significance was set at  $\alpha = 0.05$ .

## RESULTS

Between 2014 and 2019, 3484 kratom-related exposures among adults were reported to U.S. poison centers;

162 cases (4.6%) were among adults over 60 years. The number of kratom-related exposure cases per year among all adults increased over the study period (Figure 1). Across age groups, the annual number of cases per year was greater among younger adults relative to adults in each older age group ( $p < 0.01$ ).

The majority of cases originated with healthcare facility calls (81.1%), such as hospitals, emergency departments, or physician's offices. (Healthcare facilities do not include nursing homes or other chronic care residential facilities.<sup>32</sup>) Overall, cases predominantly involved kratom as a single ingestant (63.0%); multiple ingestants were significantly less common among older adult cases ( $p = 0.01$ , Table 1). Most cases (70.4%) were related to acute (single episode of use) exposures. Among older adults, half of the cases were among women, compared with 30.8% of cases among adults aged 18–59 years ( $p < 0.001$ ). Reasons for ingestion were most often intentional exposures, primarily abuse (use of the substance to gain a psychotropic effect) or misuse (use of a substance for reasons other than a psychotropic effect) (58.2%, 50.8%, and 43.8% among adults aged 18–59, 60–69, and

70 years and above, respectively), and suspected suicide (11.4%, 8.5%, and 15.6% among adults aged 18–59, 60–69, and 70 years and above, respectively). Unintentional exposures and adverse reactions were less common overall, but were significantly more common among older adults ( $p = 0.03$ ), with 21.9% of cases among ages 70 and older involving an adverse reaction (e.g., drug interaction with other medications), compared with 9.6% among ages 18–59 and 12.3% among ages 60–69 years. Across all cases, kratom was reportedly consumed in solid (tablet/capsule, 39.8%), powder/granule (15.2%), or liquid (7.5%) forms and exposure routes were ingestion (56.9%), inhalation (2.0%), and ingestion with inhalation (2.0%), although the route of exposure was unknown for many cases (39.0%) (Table 1).

A total of 51 deaths and 404 cases with major effects were observed among all cases (3 and 20 among adults 60 and older, respectively); among cases where kratom was the only substance involved there were 8 deaths and 207 cases with major effects (data not shown). Approximately half of the cases where kratom was the only substance involved resulted in a serious clinical outcome

| Medical outcome <sup>a</sup>                                                                            | 18–59 years<br>( <i>n</i> = 2076) | 60–69 years ( <i>n</i> = 97) | 70 years and above<br>( <i>n</i> = 97) | All cases ( <i>N</i> = 2196) | <i>p</i> -value   |
|---------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------|----------------------------------------|------------------------------|-------------------|
| No effect                                                                                               | 5.1                               | 4.1                          | 0.0                                    | 5.0                          | 0.08 <sup>b</sup> |
| Minor effect                                                                                            | 26.5                              | 30.9                         | 13.0                                   | 26.6                         |                   |
| Moderate effect                                                                                         | 39.6                              | 34.0                         | 52.2                                   | 39.5                         |                   |
| Major effect                                                                                            | 9.3                               | 10.3                         | 17.4                                   | 9.4                          |                   |
| Death <sup>c</sup>                                                                                      | 0.3                               | 1.0                          | 0.0                                    | 0.4                          |                   |
| Not followed (at most minimal effects)                                                                  | 8.3                               | 13.4                         | 4.3                                    | 8.5                          |                   |
| Unable to follow (potentially toxic exposure)                                                           | 10.8                              | 6.2                          | 13.0                                   | 10.7                         |                   |
| Clinical effects <sup>a,d</sup>                                                                         | 18–59 years<br>( <i>n</i> = 2076) | 60–69 years ( <i>n</i> = 97) | 70 years and above<br>( <i>n</i> = 23) | All cases ( <i>N</i> = 2196) |                   |
| Neurological (e.g., agitation, coma, confusion, dizziness, drowsiness, hallucinations, seizure, tremor) | 75.6                              | 67.0                         | 85.2                                   | 75.4                         | 0.06              |
| Cardiovascular (e.g., hypertension, tachycardia)                                                        | 44.8                              | 36.8                         | 51.9                                   | 44.5                         | 0.15              |
| Gastrointestinal (e.g., abdominal pain, nausea, vomiting)                                               | 25.2                              | 24.5                         | 25.9                                   | 25.2                         | 0.73              |
| Respiratory (e.g., respiratory depression)                                                              | 11.8                              | 15.1                         | 22.2                                   | 12.1                         | 0.12              |

Note: Values represent percentages. *p*-value indicate results from Fisher's exact test used to compare distribution across age categories.

<sup>a</sup> Single exposure kratom cases only; cases with exposure to more than one substance omitted.

<sup>b</sup> *p*-value reflects comparison across age groups and medical outcomes; cases that were not or unable to be followed not included.

<sup>c</sup> Deaths: *n* = 7 among ages 18–59, *n* = 1 among 60–69, *n* = 0 among 70 years and above.

<sup>d</sup> Among cases with at least one symptom listed.

**FIGURE 2** Clinical effects of single-ingestant kratom exposures among adults aged 60–69 and above 70 years reported to U.S. poison centers, 2014–2019



(49.3%), including 69.6% among people over age 70 years (Figure 2). Common clinical effects were neurological (75.4%), cardiovascular (44.5%), gastrointestinal (25.2%), and respiratory (12.1%) (among cases with at least one clinical effect listed). Although the presence of clinical effects by group was not significantly associated with age group, the reported prevalence was greatest among adults aged 70 and older for all the most common types of clinical effects (Figure 2).

## DISCUSSION

The rise in kratom-related exposures among older adults observed in this study mirrors the increasing trend seen in the general population.<sup>20</sup> Although others have examined kratom-related exposures using poison center data previously,<sup>20,22,31</sup> this study details kratom-related clinical effects and outcomes specifically among older adults, a population at potentially greater risk for interaction with prescription medications.

Our study's findings support concerns about the use of kratom among older adults. Pain prevalence is increasing in the older adult population.<sup>33</sup> With increasing awareness of the risks associated with opioid medications, and therefore decreased prescribing, older adults may be motivated to seek alternatives to opioids for pain control and are encouraged to do so by statements from peers and users that kratom products are safe and accessible.<sup>34</sup> Kratom is perceived as an opioid substitute to stop or reduce opioid use<sup>19</sup>; however, recommendations to use kratom and support for its potential therapeutic value must be tempered by the awareness that its safety and effectiveness have not been thoroughly evaluated by the regulatory authorities in the United States.<sup>34</sup>

Although older adults do not comprise the largest proportion of people who use kratom, their use of this substance may be particularly concerning. First, there are age-related changes in pharmacokinetics that affect drug absorption, distribution, metabolism, and elimination.<sup>35</sup> Because older adults, such as those over age 60 years, experience increased pharmacodynamic sensitivity to psychoactive effects compared with younger adults, an equivalent dose of a psychoactive medication can result in more pronounced effects as well as an increased risk for potentially harmful side-effects.<sup>36</sup>

Proportionally more kratom-related cases among older adults (aged 70 and above) involved adverse reactions or unintentional exposures compared with younger adults. According to national estimates from 2013 to 2016, 88.5% of U.S. adults aged 65 and older used at least one prescription medication in the last month; 39.4% use at least five prescription medications.<sup>37</sup> Prescription drug

use among older adults is also increasing over time,<sup>37</sup> as is dietary supplement use.<sup>38</sup> The prevalence of concurrent use of prescription drugs and herbal medications among older adults ranges from 45.3% to 88.3% in the United States.<sup>39</sup> Use of kratom, like other herbal medications, should be monitored and managed by a healthcare provider to mitigate potential interactions.<sup>39,40</sup>

Other factors related to kratom exposures among older adults also merit consideration. Relative to younger adults, proportionally more adults aged 60 and older in this study were women, a potentially concerning finding because female sex is a known factor in adverse drug reactions.<sup>41</sup> Older adult women using psychoactive medications alone are vulnerable to a number of different injuries that contribute to morbidity and mortality, including falls.<sup>42</sup> Although the U.S. Preventive Services Task Force advises multifactorial approaches to preventing falls,<sup>43</sup> including screening for use of substance that may contribute to falls, kratom is unlikely to be listed on standard medication histories used in risk assessment. Online studies using social media found that 40% of adults aged 31–50 years share information about kratom use with healthcare professionals, however, analogous studies among older adults are lacking.<sup>7</sup>

Our findings confirm those from prior studies associating kratom exposures with outcomes that require medical intervention. A previously published study reported that from 2011 to 2017, 1807 kratom exposures were reported at U.S. poison control centers, half of which (51.9%) resulted in a serious medical outcome.<sup>31</sup> While deaths associated with kratom are not common, they have been reported.<sup>44–46</sup> A recent analysis of unintentional drug reporting data from 32 states identified kratom as a cause of 91 deaths and involved in another 61 deaths between July 2016 and December 2017.<sup>47</sup> Although other drugs were detected in most of these deaths, in seven cases kratom was the only detected substance.<sup>47</sup> Similarly, in this study, two-thirds of exposures involved kratom as a single substance, among which about half of all exposures required significant medical intervention, and there were some deaths. Our study further showed that more than two-thirds of the exposures among older adults required significant medical intervention.

Some federal actions have been taken in response to concerns about kratom. Regulatory agencies have expressed concerns about kratom as early as 2016.<sup>48</sup> In November 2017, the FDA issued a public health advisory warning consumers not to use kratom due to its potential for addiction and abuse, as well as other safety concerns.<sup>49</sup> More recently, in June 2019, the FDA banned kratom imports<sup>50</sup> and issued warning letters to two companies that market and distribute kratom, citing unsubstantiated claims that advertise the use of kratom

for opioid cessation, pain management, or other medical uses.<sup>51</sup> Some states and localities have banned kratom and variations of the Kratom Consumer Protection Act have been passed in Utah, Georgia, and Arizona in 2019, which requires safety testing and labeling standards for kratom products.<sup>52</sup> If use and corresponding health outcomes associated with kratom exposure continue to increase, more regulatory action may be warranted.

This study has limitations. First, the NPDS captures exposure cases reported to poison centers in the United States, which may involve cases with more obvious or serious clinical effects requiring the assistance of a poison center or toxicologist and may be an underrepresentation of the true number of significant cases. These data are not equivalent to a primary medical record review and may not contain as detailed data as would be in a medical record of exposed individuals. Deaths may also be unverifiable, lost to follow-up, or undercounted. Clinical effects reported to poison centers may be attributable to other medications or substances, including those not disclosed by the patient or detected in urine drug screens. Poison center specialists classify exposure substances, details, and clinical effects over the telephone; factual inaccuracies may therefore arise in the course of data collection. Confirmatory assays and testing for kratom are not necessarily utilized to verify exposure.

## CONCLUSIONS

This study illustrates increasing adverse events attributed to kratom use which generated contact with poison centers in the United States. Older adult patients are using kratom, presumably concurrent with other medications and supplements. Healthcare providers and patients should be aware of the potential for adverse effects, some of which could be serious. By discussing kratom use and potential toxicology with patients, healthcare professionals can help patients disclose needs and risk which may be otherwise unassessed. Improving opportunities to discuss kratom and other substance use with all patients will contribute to optimizing health as well as the ability to provide safe alternatives for patient symptom relief.

With the increasing access to and use of kratom in the United States, future research is needed on patterns of use, and co-use with other medications, particularly among older adults. Future research should track adverse events to better understand the potential for benefits and harms associated with kratom and to determine whether regulation of kratom may be needed to protect public safety, including among older adults.

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## CONFLICT OF INTEREST

The authors have no conflicts of interest to report.

## AUTHOR CONTRIBUTIONS

Dr Janessa M. Graves conceptualized the study, drafted the initial manuscript, and coordinated the review and revision of the manuscript. Dr Julia A. Dilley assisted in the design of the study, carried out analyses, and provided critical review and revision of the manuscript. Drs Ashley Brooks-Russell, Jennifer M. Whitehill, Tracy A. Klein, and Erica Liebelt, and Ms Lucia Terpak assisted with the analysis plans, interpretation of findings, and critical review and revision of the manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

## SPONSOR'S ROLE

The sponsor had no role in the design, methods, subject recruitment, data collections, analysis, and preparation of the article.

## DISCLAIMER

The American Association of Poison Control Centers (AAPCC; <http://www.aapcc.org/>) maintains the national database of information logged by the country's poison centers (PCs). Case records in this database are from self-reported calls: they reflect only information provided when the public or healthcare professionals report an actual or potential exposure to a substance (e.g., an ingestion, inhalation, or topical exposure), or request information/educational materials. Exposures do not necessarily represent poisoning or overdose. The AAPCC is not able to completely verify the accuracy of every report made to member centers. Additional exposures may go unreported to PCs and data referenced from the AAPCC should not be construed to represent the complete incidence of national exposures to any substance(s).

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### Editor's Note

Graves and her colleagues call our attention to an unregulated herbal product, kratom (*Mitragyna speciosa*). Historically, the dried herb was chewed as a stimulant to reduce fatigue and increase heat tolerance by manual laborers in Southeast Asia. Current uses include fatigue, depression, and as an opioid alternative. The FDA has stated the kratom has no approved uses, and advised consumers and clinicians against its use due to safety concerns. One concern for older adults taking kratom is potential drug–drug interactions. Mitragynine undergoes phase 1 (hydrolysis) and 2 metabolism (glucuronidation). Studies suggest that kratom inhibits CYP3A4, 2D6, and 1A2 isozymes. Concomitant use with carisoprodol, modafinil, fentanyl, diphenhydramine, morphine, O-desmethyltramadol, and others are reported to have resulted in serious and fatal adverse effects. Clinicians should ask their patients if they are taking kratom or kratom-containing products, and advise them against such use.

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